Basic Principles of Medicine 1
Module: Cardiovascular, Pulmonary, Renal
DLA 14

Lung Function and Respiratory Gases 2

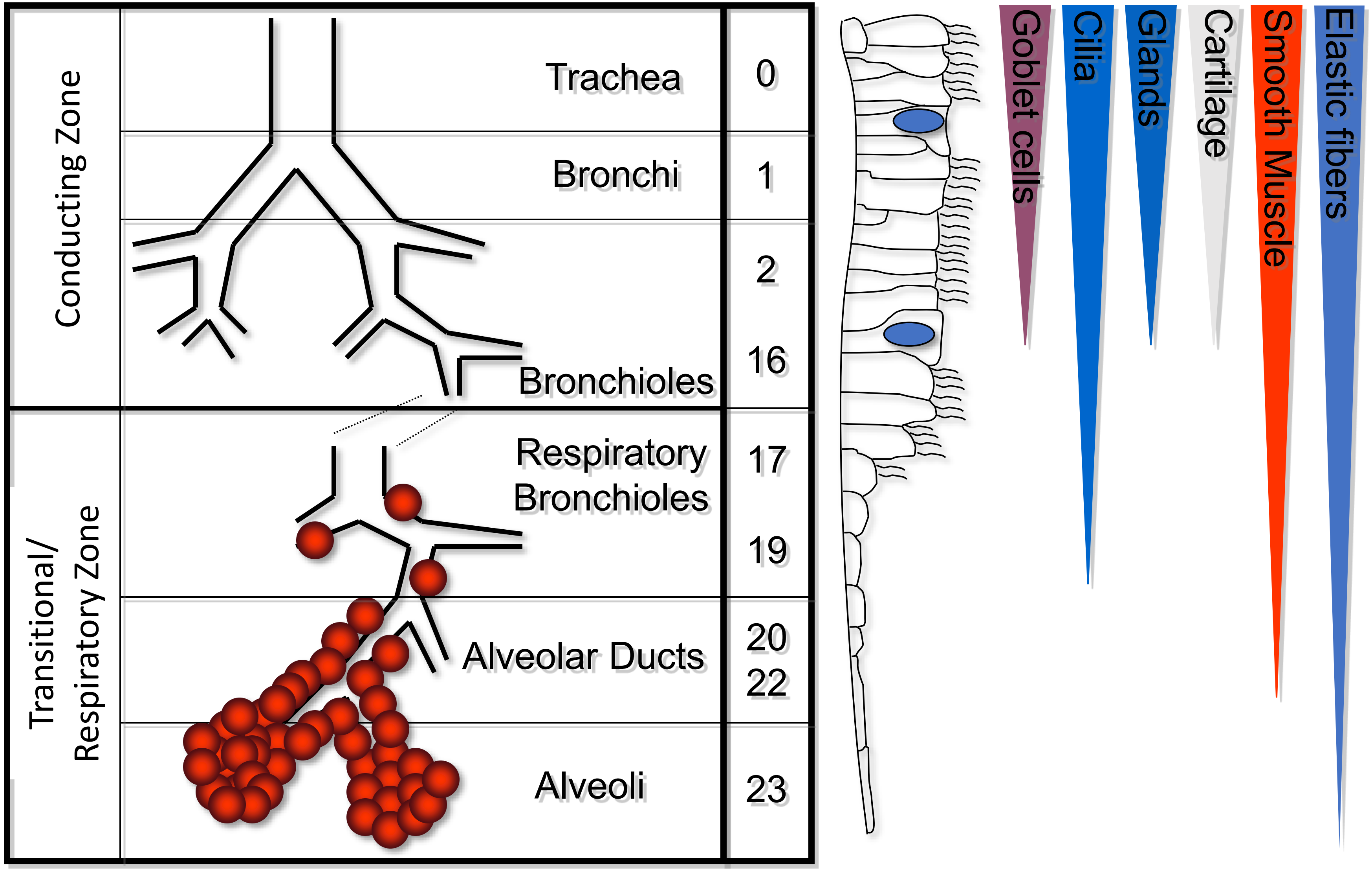
Dr. Juanette McKenzie
St Georges University ©

Learning Objectives

By the end pf this DLA, the student should be able to

SOM.MK.II.BPM1.3.CPR.2.PHYS.0191	Describe the structural characteristics of airways.
SOM.MK.II.BPM1.3.CPR.2.PHYS.0192	Describe the transport of gas through the conducting airways to and from the alveoli.
SOM.MK.II.BPM1.3.CPR.2.PHYS.0193	Explain the key roles of mucociliary transport and describe related pathologies and their clinical presentation.
SOM.MK.II.BPM1.3.CPR.2.PHYS.0194	Describe the clinical significance of mucociliary transport.
SOM.MK.II.BPM1.3.CPR.2.PHYS.0195	Describe the functions of alveolar Type 1 & type II cells.
SOM.MK.II.BPM1.3.CPR.2.PHYS.0196	Lists components of chest wall and lungs and relate the functions of respiratory muscles to movement of air in and out of the alveoli.
SOM.MK.II.BPM1.3.CPR.2.PHYS.0197	Describe how Boyle's and Charle's law apply to movement of gas in and out of lungs.
SOM.MK.II.BPM1.3.CPR.2.PHYS.0198	List the metabolic functions of the lungs and explain their key physiological roles.

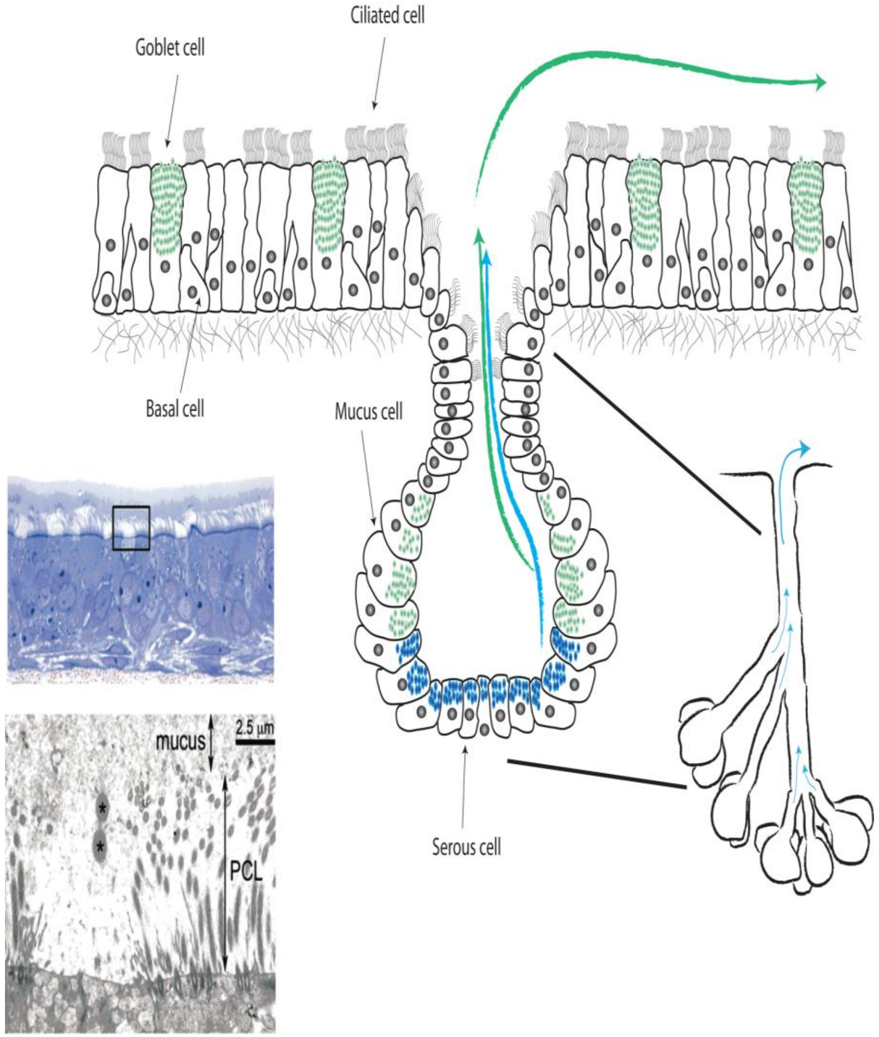
Respiratory system – Airway Structure



Respiratory System – Mucocilary Transport

•The cilia lining the airways beat the mucus covering them away from the alveoli, and toward the pharynx (**mucociliary escalator**)

•Studies show that ciliary function is inhibited or impaired by cigarette smoke.



Diseases of Conduction Zone & Respiratory Zone

Zone	Function	Structures	Pathology	Examples
Conducting	Airflow	Airways	Ventilation Problems – Flow velocity problems	Asthma, COPD, bronchitis, bronchiolitis, bronchiectasis, cystic fibrosis
Respiratory	Gas Exchange	Alveoli/Lung Parenchyma	Gas exchange Problems – may present with volume problems	Pulmonary Fibrosis (including pneumoconiosis) , hypersensitivity pneumonitis, pneumonia,

Clinical Consideration – Chronic Bronchitis

- Inflammation of the bronchi
- Commonly caused by cigarette smoking
- The number of goblet cells may increase and the mucous glands may hypertrophy
- Cilia movement often is impeded.
- Increased mucous gland secretion and increased viscosity of mucus, leads to coughing and obstruction

Cough with sputum expectoration for at least 3 months a year during a period of 2 consecutive years.

Summary

- The mucociliary transport system of the respiratory tract is a protective mechanism.
- Injury to the mucociliary transport mechanism can lead to problems in the conduction in airflow.